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Shortening the albumin challenge from 48 to 24 hours may lead to overdiagnosis of hepatorenal syndrome-acute kidney injury and overtreatment with terlipressin

To the Editor:
The definition of hepatorenal syndrome-acute kidney injury (HRS-AKI) has undergone multiple transformations since the first iteration in 1979.[1] The recent Acute Disease Quality initiative (ADQI)-International Club of Ascites (ICA) consensus paper[2] has revised the prior 2015 definition.[3] One of the major proposed changes relates to the use of albumin. While the 2015 definition included the absence of AKI response to a 2-day albumin challenge of 1 g/kg/day, the updated definition has abbreviated the use of albumin (or crystalloids), by requiring only "the absence of improvement in serum creatinine and/or urine output within 24 h following adequate volume resuscitation (when clinically indicated)". This new definition relies on individual clinician's assessment of what is considered adequate volume and when it is indicated. Of note, although the previously recommended use of albumin in AKI, including amount and duration, was mostly based on expert opinion and on data in other settings, such as spontaneous bacterial peritonitis and large volume paracentesis,[4,5] the new recommendation is also based on expert opinion. As such, there is no evidence to suggest that the 24 h treatment is advantageous over 48 h of treatment.

In this regard, our group recently reported the results of a prospective cohort of hospitalized patients with cirrhosis and AKI in which we evaluated the efficacy and safety of the different steps in the management algorithm of AKI proposed by EASL.[6] One of the issues we evaluated was the efficacy of albumin administration. We found that 47 of the 139 (34%) patients with AKI stage ≥1B treated with albumin during the first 48 h from AKI diagnosis achieved response to albumin, defined as downstage to AKI 1A or resolution of AKI (a return of serum creatinine to within 0.3 mg/dl of baseline). This is a surprisingly high rate of AKI response to a relatively simple therapeutic intervention.

To shed some light on the controversy between 24 h *vs*. 48 h albumin challenge, we reviewed these 47 patients with AKI stage ≥1B who responded to albumin at 48 h. All patients but one had serum creatinine values at 24 h and at 48 h after the start of the albumin administration. Of these 46 patients, 28 (60.9%) achieved response at 24 h, while the remaining 18 patients (39.1%) required 48 h to achieve response. Only one patient had progression of AKI stage at 24 h prior to responding to albumin at 48 h. Patients with response at 24 h did not differ with respect to baseline characteristics from those without response at 24 h.

In brief, almost 40% of patients who respond to albumin by 48 h do not show early response by 24 h. With the new ADQI-ICA consensus definition, these patients could have been labelled as having HRS-AKI and been treated with vasopressors such as terlipressin, potentially exposing them to important side effects, when in fact another 24 h of albumin administration would have been sufficient. At the time of developing the ADQI-ICA position paper, we had in fact not yet published our results on AKI response to albumin. Of course, there is a need for studies addressing the important questions on the dose and duration of albumin treatment to confirm the diagnosis of HRS-AKI in patients with cirrhosis. However, while awaiting these studies, our findings suggest that the duration of albumin therapy should be 48 h, unless there is early resolution of AKI at 24 h, because no further treatment is needed. While a one-size-fits-all definition for HRS-AKI may not be the solution, using a definition with ambiguity also makes identification of this syndrome more challenging and may lead to its overdiagnosis.

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Conflict of interest

PG has received research funding from Gilead and Grifols. PG has consulted or attended advisory boards for Gilead, RallyBio, SeaBeLife, Merck, Sharp and Dohme (MSD), Ocelot Bio, Behring, Roche Diagnostics International and Boehringer Ingelheim, and received speaking fees from Pfizer. The other authors have declared no conflict of interest.

Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

All authors contributed equally to the writing, reviewing, and editing of the manuscript.

Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jhep.2024.07.015.

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